

Geometry

Unit 11

Lesson 8 Homework

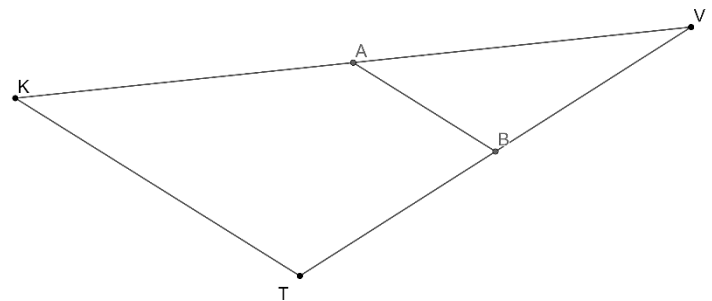
Name _____

Date _____

Period _____

Use this information to answer questions 1 through 5.

Triangle KVT is shown, where $\overline{AB}$ is the midsegment. $KT = 2x + 4$, $AB = \frac{1}{2}x + 3$, $AV = \frac{5}{2}y - 6.5$, $KA = \frac{1}{2}y + 3.5$, $VT = 3z - 7$, and $TB = 2x + 2.1$.



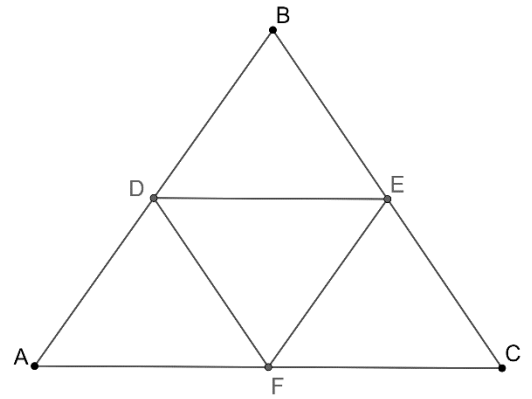
1. Find the value of x .
2. Find the value of y .
3. Find the value of z .
4. Find the length of the midsegment, $\overline{AB}$.
5. Find the length of $\overline{KV}$.

Use this information to answer questions 6 through 10.

Triangle ABC is shown, where D , E , and F are midpoints.

$DE = 2x - 4$, $AC = 4x - 8$, $DF = x + 3$, $BC = 3x + 1$,

$FE = 3x - 6$, and $AB = 5x - 7$.



6. Find the value of x .

7. Find the length of $\overline{DF}$.

8. Find the length of $\overline{EF}$.

9. Find the length of $\overline{DE}$.

10. Find the perimeter of triangle ABC .